Akoizo Original Practice - Form 4

Division B | Hovercraft | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which battery type is prohibited for powering the motors?

Unit: The electrical system: batteries, motors, and speed control.
Worked-case data: A team wants to use eight 1.5 V alkaline AA cells. Is that legal, and what are the options?
Separate investigation observations:
Monday: Fresh pack 5.2 V, knob at 3: 11.8 s.
Wednesday: Same knob, pack at 4.7 V: 13.4 s.
Wednesday, recharged: Pack 5.2 V, knob at 3: 11.9 s.

A. Nickel-metal hydride

B. Nickel-cadmium

C. Lithium-ion

D. Alkaline

Answer: _____

2. (2 points) A hovercraft drifts toward its left rail while hovering in place. A likely cause is:

A. Battery voltage too high

B. The dowel is too tall

C. More air leaking on the left side of the skirt

D. Too much nickel mass on the right

Answer: _____

3. (2 points) Which action is allowed on the track during competition time?

Unit: Impound, inspection, and competition day.
Worked-case data: A team's vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

A. Dry-sweeping it

B. Wetting it

C. Asking the supervisor to re-level it

D. Applying a slick coating

Answer: _____

4. (2 points) Where should the loaded center of mass sit?

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

A. Over the center of the air cushion

B. As far forward as possible

C. Directly over the thrust fan

D. Near one rail

Answer: _____

5. (4 points) Explain how a fan produces forward thrust, using Newton's third law.

Answer: _____
_____
_____

6. (4 points) Why can carrying more nickels lower your score even though Mass Score rises?

Answer: _____
_____
_____

7. (4 points) Setting 6 gives 14.0 s and setting 8 gives 18.0 s. Estimate a setting for 15.0 s.

Answer: _____
_____
_____

8. (4 points) State two safety rules a vehicle must pass to run.

Unit: The electrical system: batteries, motors, and speed control.
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Separate investigation observations:
Monday: Fresh pack 5.2 V, knob at 3: 11.8 s.
Wednesday: Same knob, pack at 4.7 V: 13.4 s.
Wednesday, recharged: Pack 5.2 V, knob at 3: 11.9 s.

Answer: _____
_____
_____

9. (4 points) How can the supervisor test whether a vehicle is levitating?

Unit: Levitation: the air cushion and the skirt.
Worked-case data: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?
Separate investigation observations:
Empty test: Vehicle glides freely across the floor with lift on.
With all nickels: Rear skirt drags and the craft stops quickly.
Nickels moved forward 6 cm: Glides freely again, sitting level.

Answer: _____
_____
_____

10. (4 points) How can the build log help with design-knowledge questions?

Unit: Impound, inspection, and competition day.
Worked-case data: A team’s vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.

Answer: _____
_____
_____

12. (2 points) What two variables are changing the run time here?

Supplied synthetic observations:
Trial 1: Start 5 cm behind line: 14.1 s.
Trial 2: Start 40 cm behind line: 11.6 s.
Trial 3: Start 40 cm behind, plus 2 rolls: 13.8 s.

A. Only the timer

B. Start position and carried mass

C. Battery chemistry and skirt color

Answer: _____

13. (2 points) Define “Target Time (TT)” in the context of Hovercraft.

Answer: _____
_____
_____

14. (2 points) Define “Calibration curve” in the context of Hovercraft.

Answer: _____
_____
_____

15. (2 points) Define “Parallel” in the context of Hovercraft.

Answer: _____
_____
_____

16. (2 points) Define “Skirt” in the context of Hovercraft.

Answer: _____
_____
_____

17. (2 points) Define “Competition time” in the context of Hovercraft.

Answer: _____
_____
_____

18. (2 points) Define “Cradle” in the context of Hovercraft.

Answer: _____
_____
_____